Making a sign chart recipe

A *sign chart*, as we learned in class by working on this handout, is a labeled number line. Here's an example from the textbook, also. Here's a recipe for making them:

1. Figure out all the critical numbers of $f(x)$.
   - Find the derivative $f'(x)$ and figure out where $f'(x) = 0$ or does not exist.
   - For finding zeroes, it may be useful to review factoring and the Zero Product Property.
   - For finding places where $f'(x)$ does not exist, think about what could be *bad* x-values. Maybe are there things that would make you divide by zero? Or take the square root of a negative number? Or other weird domain issues?
   - Remember that the critical numbers are *inputs*.
2. Put the critical numbers on a number line.
   - We low-key don't care about any other numbers. In particular, we low-key don't care about endpoints, at least not for now, and we probably don't care about $x = 0$, unless of course it is a critical number.
   - The critical numbers chop up the number line into a bunch of segments before and after each one.
3. Choose your favorite number from each of the segments and plug it into $f'(x)$.
   - We are making a number line about *slopes*, so we plug into $f'(x)$, not $f(x)$.
   - We low-key don't care what *number* comes out of $f'(x)$; we really only care about its *sign*: $+$ or $-$. (That's why this is called a *sign* chart!)
4. Label each segment with the *sign* of $f'(x)$ on that segment.
5. Translate the *signs* of $f'(x)$ into the *directions* – increasing or decreasing – of $f(x)$.

Finding extreme values recipe

Building on ideas from in class and on the exit ticket, here's a recipe for finding extreme values:

1. Figure out all the critical numbers of $f(x)$.
   - I am once again emphasizing that the critical numbers are *inputs*.
2. Plug the critical numbers into the original function $f(x)$.
   - The outputs that result are called the *critical values*.
   - What would happen if you accidentally plugged the critical numbers into $f'(x)$?
3. The endpoint numbers are also inputs, so plug them into the original function $f(x)$ too.
4. Compare all these original $f(x)$ outputs and find the biggest and the smallest.